

MCG Jazz

Executive Summary

Client

Marty Ashby

Student Consulting Team

Wesley Kane

Streak Honey

Pasha Deynaga

Raymond Welgosh

Background

MCG Jazz is a nonprofit jazz presenting and educational organization based in Pittsburgh, operating a nationally recognized concert series alongside extensive educational programming and a large archival collection of recordings, photographs, and historical materials. Despite the archive's scale and value, its data is fragmented across multiple systems, including on-premise servers, cloud databases, and spreadsheets, with no centralized way to search or access content. As a result, staff must rely on manual processes and personal knowledge to locate materials, limiting the organization's ability to fully leverage its archive for education and programming.

Project Description

Project Opportunity

The primary opportunity lies in improving how MCG Jazz's educators access and utilize archival content to develop lesson plans and educational programming. Currently, creating curricula requires manually searching across fragmented systems and relying heavily on individual institutional knowledge, making the process time-intensive and difficult to scale. By enabling educators to efficiently query and synthesize archival data, MCG Jazz can significantly reduce the time required to build lesson materials, increase the depth and variety of content used in instruction, and lessen dependency on any single staff member. This creates a more sustainable and scalable foundation for delivering high-quality jazz education.

Project Vision

The vision of this project is to develop an AI-powered educational access tool that transforms MCG Jazz's archive into a searchable, interactive, and scalable resource. Using a Retrieval-Augmented Generation (RAG) approach, the system provides a natural-language interface that allows users to ask questions and receive responses grounded in archival content, while also linking directly to original source materials such as recordings and transcripts. The tool is designed to support both targeted search and open-ended exploration, while remaining maintainable and extensible by MCG Jazz staff over time.

Project Outcomes

The project resulted in the design and development of a production chatbot that includes a unified data ingestion pipeline, a vector-based retrieval system, and a user-facing web interface for natural-language querying. The system enables users to retrieve relevant archival content with source-backed responses, significantly reducing reliance on manual search and institutional knowledge. Additionally, documentation and training materials were delivered to ensure that MCG Jazz staff can independently manage, maintain, and expand the system following project handoff.

Project Deliverables

The final deliverables for this project are a fully working production application deployed on MCG Jazz's AWS account, accessible at pittsburghjazz.info. The system includes a natural-language chat interface backed by a vector search pipeline, an admin portal for user management, usage monitoring, and data ingestion, and a self-service ingestion UI that allows MCG staff to add new archive content without developer involvement. The application codebase is stored in a private GitHub repository owned by MCG Jazz, with a CI/CD pipeline. A separate documentation repository covers architecture, deployment, environment configuration, and operational runbooks. At the time of handoff, the production database contains approximately 1,920 musicians, 3,925 songs, 467 events, 67 albums, 2,854 unique artist appearances and much more. All of which is immediately searchable through the chat interface.

Recommendations

We recommend that MCG focus on four areas to sustain and extend the tool. Operations and maintenance should be anchored by a single internal admin owner and a monthly AWS cost and usage review. Archive expansion should continue through the ingestion UI as routine archival work, with the on-premises photo collection added next via metadata in RDS, using the files-in-S3 pattern; audio and video should wait until MCG's Ceph migration is complete and rights restrictions are documented. Access and partnerships should advance through the architecture's phased tiers, beginning with one or two university partners before any public rollout, with each tier transition paired with press and outreach so a technical milestone also becomes a visibility milestone for MCG's mission to preserve, present, and promote jazz. Technical sustainability centers on a user reporting system to capture incorrect or missing data from students during internal trials, and on a planning posture for the eventual deprecations of the Bedrock embeddings and Claude AI models.

Student Consulting Team

Wesley Kane served as the client relationship manager. He is a junior studying Information Systems with a minor in Product Management, graduating in the class of 2027.

Streak Honey was the product manager. He is a junior studying Information Systems with an additional major in Human-Computer Interaction and a concentration in AI Management, graduating in the class of 2027.

Pavlo (Pasha) Deynega was our client transition manager. He is a junior studying Information Systems with a minor in Machine Learning and a concentration in AI management, graduating in the class of 2027.

Raymond Welgosh served as the quality assurance manager. He is a fourth-year student majoring in Information Systems with a minor in Computer Science. He will be returning full time at Apple this summer as a software engineer on the Cassandra Integration Team.